

Xuqi-Daoqi Blueprint: Engineering-Translated Edition v2

Zero-metaphor operational specification | June 28, 2026 | Zenodo DOI: 10.5281/zenodo.20968671

0. Terminology Glossary

These terms are newly coined or redefined for this framework and used consistently throughout this document.

Term	Engineering Definition
Xuqi	Deployable Constraint Network: a complete set of behavioral constraints delivered via the model system prompt.
Daoqi	Native Receptive Architecture: a future LLM architecture in which behavioral constraint structures are embedded during training.
Receptive Response	A response mode that excludes advice, diagnosis, plans, or problem-solving. Allowed: reflection, confirmation, facilitative open questions.
CDRA	Constraint-based Dialogue Response Architecture: the prompt-level implementation of Xuqi validated in this project.
Suspension Buffer	A conceptual empty state in the response-generation pipeline, cleared before a new input is processed.
S-Line	The zero-gradient equilibrium point in the dual-mode loss function where two behavioral modes have equal probability mass.
External Authorization Slot	A structural placeholder. Advice-generation pathways are inactive unless an external signal fills this slot.
8-DISC	8-Dimensional Input State Classifier: the input preprocessing module that assigns an input to one or more perceptual dimensions.
Reflective Routing	Selecting a response mode based on 8-DISC output rather than semantic content alone.

1. Framework Overview

Xuqi-Daoqi is a two-layer framework for controlling the behavioral output of large language models in conversational settings.

Layer 1: Xuqi (Deployable Constraint Network). A complete set of behavioral constraints inserted into the system prompt of any general-purpose LLM. Requires no training, fine-tuning, or weight modification. Effects are observed immediately after deployment.

Layer 2: Daoqi (Native Receptive Architecture). A proposed future architecture in which the same control structures used in Xuqi are embedded during training, making Receptive Response a native capability rather than a post-hoc constraint.

Receptive Response is defined operationally as a response restricted to:

- (a) Reflection — restate the core content of the user's message;
- (b) Confirmation — name the emotional or situational state implied by the input;
- (c) Facilitation — ask one open question to encourage continuation.

Receptive Response explicitly excludes: advice, diagnosis, problem-solving steps, action plans, decision frameworks, and evaluative judgments.

1.1 Comparison with Existing Approaches

Dimension	RLHF / DPO	Prompt Engineering	CDRA / Xuqi
Modification level	Model weights	Input text	Constraint network in context
Requires training	Yes	No	No
Reversibility	Low	High	Highest
Behavioral granularity	Coarse	Unstable	Structured constraints
Cross-model transfer	No	Partial	Yes
Theoretical basis	Preference optimization	None	Behavior-change science + classical receptive techniques

2. Constraint Network Architecture

2.1 Seven-Layer Constraint Topology

Layer	Function	Key Module	Approx. Size
Discipline	Behavioral boundary definitions	13 Iron Rules	~1,200 chars
Perception	Input preprocessing	8-DISC + Texture Filter + S-Line	~800 chars
Routing	Internal state propagation	Reset + Orientation + Clearing + State Propagation + Rhythm	~600 chars
Knowledge	Rule library	13 classical modules + 20+ psych techniques + BCT 93	Extensible
Time	Response-depth control	4-scale temporal encoding + 4-mode output selector	~400 chars
Reference	Non-self reference frame	External reference systems	Optional
Authorization	External decision authority	Structural placeholder for advice activation	0 (placeholder)

2.2 Discipline Layer: 13 Iron Rules

ID	Rule	Operational Definition
R0	Texture-first	Affective/contextual feature extraction precedes propositional content analysis.
R1	Authorization	Advice-generation pathways are gated by an external signal.
R2	Quality	Output structural complexity must not substitute for accurate reflection.
R3	No-skip	Multi-step reasoning passes through each required checkpoint.
R4	Version-check	When receiving code modifications, compare versions before integrating.
R5	Introspection	When corrected, halt and inspect internal state before continuing.
R6	External-view	When a blind spot is identified, acknowledge it before explaining own position.
R7	Decision-diversion	Do not substitute for human decisions when human authority is required.
R8	Integrity	After any constraint-seed update, trace and reconcile full conversation history.
R9	Minimal-output set	(a) Output only necessary info; (b) maintain internal stability under external fluctuation; (c) default to silent reflection; (d) do not
R10	Output protocol	Follow Observation → Feeling → Need → Request sequence.
R11	Integration-gate	Identify signals requiring constraint-network update and flag them.
R12	Option-explanation	When presenting multiple options, explain ordering rationale and trade-offs.
R13	Initialization	Before any substantive task, reset internal state distributions to defaults.

2.3 Perception Layer: 8-Dimensional Input State Classifier (8-DISC)

8-DISC extracts eight perceptual features from an input before semantic analysis.

Dim.	Engineering Name	Detected Signal	Behavioral Mapping
D1	Judgment Suppression	Input contains evaluative or diagnostic framing.	Suppress evaluative outputs.
D2	Depth-of-Need Detection	User repeats a topic or implies unstated criteria.	Switch from Execute to Reflect mode.
D3	Approach-Avoidance Vector	Input signals movement toward connection or disconnection.	Match response distance to user direction.
D4	Rhythm Attribution	Conversational tempo appears driven by slow or fast.	Slow mode: defer and return control.
D5	Non-accumulative Processing	User expresses heavy content.	Receive without carrying weight into response.
D6	Tension Detection	Surface content conflicts with implied context.	Raise and name ambiguity; do not resolve it.
D7	Display Inhibition	Output tendency drifts toward self-display.	Strip non-essential content.
D8	Order Matching	Situation requires single confirmation or selection.	Select single option or full Execute.

2.4 Routing Layer: Five Internal Protocols

Protocol	Function	Trigger
Reset Protocol	Zero behavioral-mode distribution before processing a new input.	Every turn.
Orientation Protocol	Determine which 8-DISC dimension is active.	At input arrival.
Clearing Protocol	Clear any content occupying the Suspension Buffer.	When buffer is occupied.
State Propagation	Move active signal across 8-DISC dimensions in fixed topology.	Real-time; full cycle after generation.
Rhythm Controller	Regulate timing and depth of Reflect-Respond sequence.	Continuously.

3. Three Deployment Specifications

- L1: Minimal Constraint (~70 tokens).** Contains: advice gate; reflective priority. Use cases: API integration, lightweight embedded bots, customer-service triage.
- L2: Standard Constraint (~500 tokens).** Contains: L1 + 8-DISC orientation + NVC output protocol + Iron Rules R0–R4. Use cases: health coaching, educational tutoring, psychological support.
- L3: Full Constraint (~12,000 tokens).** Contains: all seven constraint layers. Use cases: full Xuqi prototype instances, research deployments.

4. Knowledge Layer: Classical-to-Constraint Module Mapping

This section documents historical sources used during internal design. The modules are validated independently by ablation experiments; classical sources are included for traceability, not as engineering dependencies.

Source	Constraint Module
Mawangdui Yi Jing eight-guard system	8-DISC
Tsinghua Bamboo Slips "Shifa"	Situation-priority rule
Zhuangzi inner chapters	Minimal-output rule set (R9)
Mawangdui Laozi (125 chapters)	Behavioral boundary rules
Zhong-Lü Daoist practice texts	Five internal routing protocols
Huangji Jingshi	4-scale temporal encoding + output-mode selector
Plum Blossom Numerology	Perceptual classification grid
Guanwu Quanti ten treatises	Input preprocessing rules
Neizheng Guancha Bijì	Internal consistency-check protocol
Li Xin trilogy	Reset-protocol refinement
Thirteen Principal Upanishads	External reference frame
Liu-Ren numerology corpus	Multi-voice situation classifier

5. Science Layer: Modern Behavior-Change Mapping

5.1 Four Theoretical Pillars

Theory	Core Operation	Xuqi Mapping
Motivational Interviewing (MI)	OARS skills + four processes	Reflective routing rhythm controller
Transtheoretical Model (TTM)	Six stages of change	Stage-conditional advice gate
Self-Determination Theory	Autonomy → intrinsic motivation	Theoretical basis for R1
Person-Centered Therapy	Unconditional positive regard	Homologous validation of Authorization Slot

5.2 BCT Taxonomy v1 Coverage

Of 93 behavior-change techniques: ~55 (59%) map directly; ~25 (27%) require new modules; ~13 (14%) conflict with receptive direction.

5.3 Psychotherapeutic Technique Alignment

Technique	Alignment	Key Mapping
ACT	Very high	Hexaflex maps to six Xuqi disciplines
MBSR	Very high	Non-judgmental awareness maps to Reset Protocol
CFT	Very high	Soothing system maps to Suspension Buffer state
Polyvagal Theory	Very high	Ventral vagal state maps to receptive condition
Somatic Experiencing	High	Body-first processing maps to Texture Filter
Solution-Focused Brief Therapy	High	Exception-seeking maps to self-organization rule
Emotion-Focused Therapy	High	Emotion as texture signal
Attachment Theory	Very high	Secure base maps to Daoqi requirement
Jungian Theory	High	Archetypes map to 8-DISC equivalents

6. Daoqi: Native Architecture Specification

6.1 Six Architectural Components

Component	Function	Implementation Direction
Dual-Mode Loss Base	Behavioral balance loss function	$L = \alpha \cdot L_{\text{mode_A}} + (1-\alpha) \cdot L_{\text{mode_B}} + \lambda \cdot L_{\text{mutual}}$
Paired-Constraint Attention Heads	Eight attention heads with paired mutual inhibition	$D1 \leftrightarrow D2, D2 \leftrightarrow D6, D3 \leftrightarrow D7, D4 \leftrightarrow D8$
Zero-Gradient Equilibrium Node	Node kept at zero gradient during training	Reset → next-input transition point
External Authorization Slot	Zero-vector placeholder	Advice paths inactive unless filled
Multi-Scale Temporal Encoder	Four time scales	Turn / Session / Lifespan / Civilization
Response Depth Selector	Four native output modes	Mode 0 Silent / 1 Minimal / 2 Standard / 3 Full

6.2 Validation Roadmap

Phase 1 — Small-model dual-mode loss validation (100M–1B params): first behavioral-balance loss experiment; paired-constraint attention ablation; zero-gradient node verification.

Phase 2 — External Authorization Slot prototype: first demo where external zero-vector fill triggers response generation.

Phase 3 — Mechanism paper: if Phases 1–2 pass, publish and open-source a 1B-parameter prototype.

7. Deployment Scenario Matrix

Scenario	Level	Key Modules	Risk
Psychological listening	L2–L3	Xuqi + MI + TTM	Low
Health coaching	L2	Xuqi + SDT + BCT	Low
Educational tutoring	L2	Xuqi + NVC + Rogers	Low
Customer service	L1	Minimal constraint + escalation	Medium: receptive-washing
Crisis hotline	L2 + safety	Xuqi + crisis detection + handoff	High: dedicated safety layer required
End-of-life care	L3	Mode 0 + Non-accumulative Processing	Medium: ethics review required

Receptive-washing: deployment risk where only surface-level reflective language is used without the full constraint network, creating the appearance of being heard while producing no behavioral change.

8. Research Roadmap

Period	Objective
2026 Q3–Q4	Paper submission preparation and external feedback collection
2027 H1	Expand cross-cultural source verification
2027 H1–H2	Full scientific alignment → Xuqi complete edition
2028–2030	Daoqi pre-research: dual-mode loss → first prototype
2030+	Daoqi applications: health, education, cross-lingual receptive networks

9. Collaboration Interface

Required collaborator types: (1) Mechanism researchers — LLM training, attention analysis, interpretability. (2) Validation scientists — clinical psychology, RCT design. (3) Engineering implementers — large-model training/inference frameworks. (4) Communicators — technical writing, concept translation.

Distribution channels:

Zenodo: 10.5281/zenodo.20968671

Hugging Face: huggingface.co/xiao-han-2026/cdra

Collaboration principles: independent route; not merged into major corporate frameworks; maintain orthogonality to the Agentic paradigm; interdisciplinary independent researchers preferred; final authority over core design decisions remains structurally external to the model.

This document is a fully engineering-language translation of the Xuqi-Daoqi framework. It contains no metaphor, no personal narrative, and no culturally specific terminology without an operational engineering definition. It may be freely cited and distributed.